



Problem statement:-

Write a program to check whether the entered number is Armstrong or not.

Theory:-

General definition:-

The number for which the sum of its digit to power of number of its digit is equal to that number is called Armstrong number.

Example:- 1: 153

Total digit in 153 is 3.

$$\text{And } 1^3 + 5^3 + 3^3 = 1 + 125 + 27 = 153$$

Example 2: 1634

Total digits in 1634 is 4.

$$\text{And } 1^4 + 6^4 + 3^4 + 4^4 = 1 + 1296 + 81 + 64 = 1634.$$

Examples of Armstrong numbers: 1, 2, 3, 4, 5, 6, 7, 8, 9, 153, 370, 371, 407, 1634, 8208, 9474, 54748, 92727, 548834, 1741725.

while loop in C.

while loop is also known as pre-tested loop.

In general, a while loop allows the part of the code to be executed multiple times depending upon a given boolean condition.

It can be viewed as repeating if statement. The while loop is mostly used in the case where the number of iterations is not known in advance.

Algorithm:-

Step 1:- Start

Step 2:- Declare variables

num, original num, rem, sum = 0

Step 3:- Input a number from the user and store it in num.

Step 4:- Store original value.

original num = num.

Step 5:- Repeat while num > 0

rem = num % 10.

sum = sum + (rem * rem * rem).

num = num / 10.

Step 6:- Check condition

if sum == original sum,

Print "Number is Armstrong"

else

Print "Number is not Armstrong"

Step 7:- Stop

ser and store it in



Syntax of while loop in C language
The syntax of while loop in C language is given below

```
while (condition) {  
    // code to be executed  
}
```

Properties of while loop:

- A conditional expression is used to check the conditions. The statements defined inside the while loop will repeatedly execute ~~while~~ until the given condition fails.
- The condition will be true if it returns 0, The condition will be false if it returns any non-zero number.
- In while loop, the condition expression is compulsory.
- Running a while loop without a body is possible.
- We can have more than one conditional expression in while loop.
- If the loop body ^{contains} only one statement, then the braces are optional.

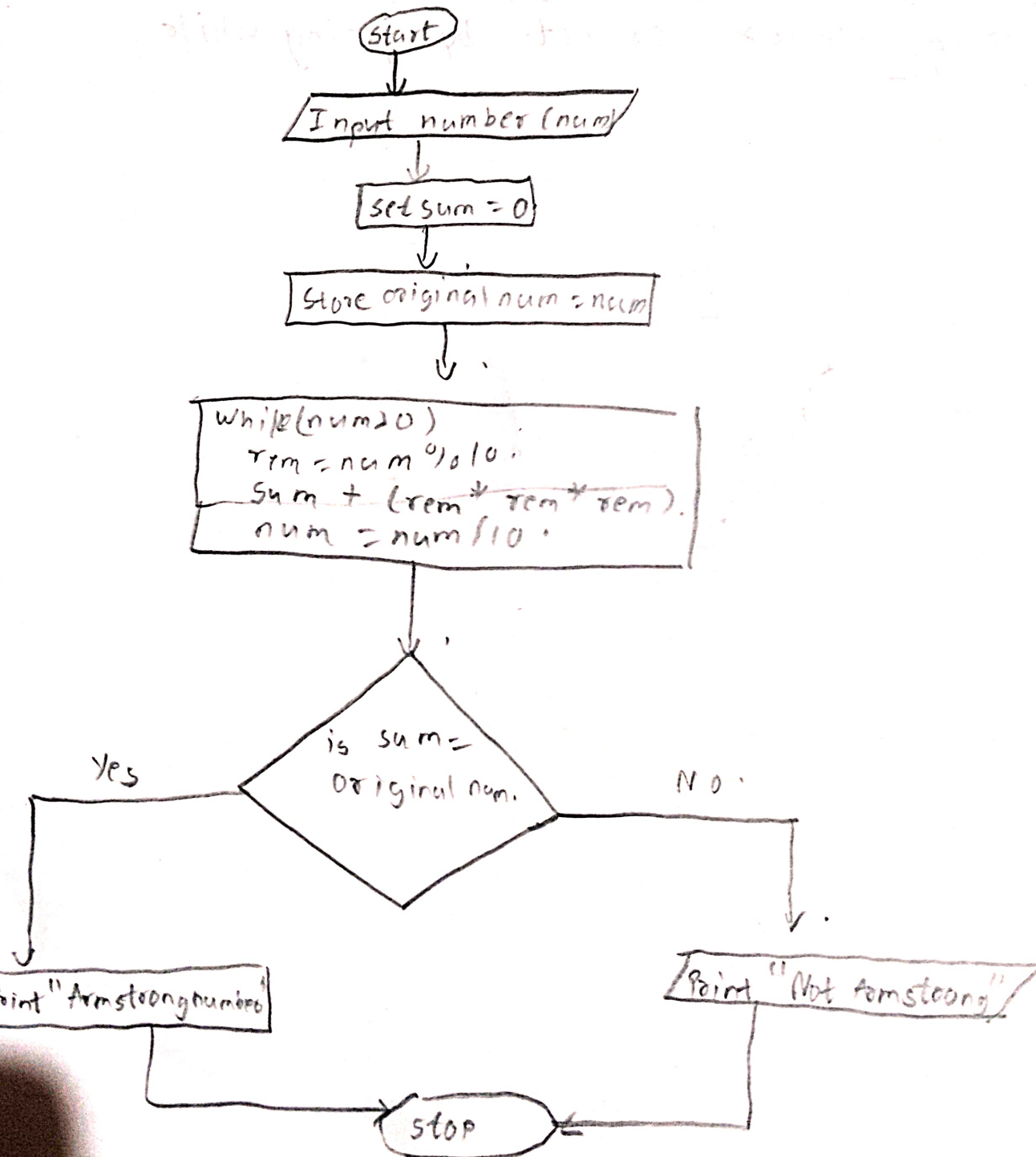


Conclusion:- We have successfully written the program to check whether number is Armstrong number or not by using while loop.

Input:-

```
#include <stdio.h>
#include <conio.h>
#include <math.h>
int main()
{
    int r, temp, num, sum, count;
    clrscr();
    printf("ENTER THE NUMBER");
    scanf("%d", &num);
    count = 0;
    temp = num;
    while (num != 0)
    {
        num = num / 10;
        count++;
    }
    num = temp;
    sum = 0;
    while (num != 0)
    {
        r = num % 10;
        sum = sum + pow(r, count);
        num = num / 10;
    }
    if (temp == sum)
        printf("The GIVEN NUMBER IS ");
    else
        printf("The Number is not Armstrong");
    getch();
    return 0;
}
```


Flowchart:-



Flowchart

